A Two-Level Multigrid Method for the Reaction–Diffusion Master Equation

Aditya Dendukuri

Abstract

We develop a two-level multigrid finite state projection (FSP) method for the reaction–diffusion master equation (RDME). The central observation is that pairwise voxel aggregation is a Kemeny–Snell lumping of the Markov chain generated by fast intra-pair diffusion. This yields a coarse generator of the Galerkin form $\bar{A} = \mathcal{R}A\mathcal{P}$, where restriction is exact marginalization and prolongation is induced by the local stationary law of the fast diffusion operator. For pair aggregation, this prolongation is Binomial.

The multigrid cycle uses intra-pair diffusion as the smoother, exact restriction to transfer the pre-smoothed distribution to the coarse level, a coarse RDME solve, and prolongation of the coarse *correction* $\delta^{2h} = \tilde{p}^{2h} - p^{2h}$ back to the fine level. This is a full-approximation scheme (FAS) for the probability generator. The coarse generator is exact for linear birth–death–diffusion systems and provides a tractable closure for nonlinear reaction networks.

Uniform coarsening fails at bistable interfaces because the Binomial prolongation cannot distinguish the two orientations of an interface pair. This motivates an adaptive extension in which the refinement criterion, based on within-pair asymmetry, inter-pair gradient, and per-molecule flux diagnostics, localizes coarsening to homogeneous bulk regions. Fine and coarse pairs are then co-evolved in a persistent mixed-resolution FSP that avoids rematerializing the full fine grid.

Numerical experiments on one-dimensional birth–death and Schlögl reaction–diffusion systems demonstrate: (i) exact coarse evolution for linear networks; (ii) V-cycle accuracy on nonlinear bistable fronts with measurable stiffness reduction; (iii) automatic front localization and persistent mixed-resolution evolution; (iv) second-level quad-coarsening with policy verification that bulk-only promotion is enforced. These results establish adaptive multigrid FSP as a distribution-resolving method for low-copy spatial systems with localized metastable structure.

1 Introduction

The reaction–diffusion master equation (RDME) provides a mesoscopic Markov description of spatial stochastic chemical kinetics by coupling local reaction channels with diffusive transport between neighboring voxels [5, 6]. When the underlying process is low-copy, multimodal, or metastable, the full probability distribution may be the object of interest rather than only moments or sample paths. Finite state projection (FSP) methods are attractive in this regime because they approximate the chemical master equation directly on a truncated state space with explicit mass control [8]. The difficulty is that spatial discretization makes the state space grow rapidly with the number of voxels, while diffusion introduces fast modes that aggravate stiffness.

This paper develops a multigrid reduction strategy for the RDME that targets the spatial dimension directly. The basic coarsening step merges adjacent voxels into pairs and formulates the resulting operator as a genuine Galerkin coarse generator. At the algorithmic level, this yields a two-level FAS-style V-cycle:

1. **Pre-smooth**: evolve the fast intra-pair diffusion modes to drive within-pair distributions toward the local equilibrium (Binomial for identical voxels).
2. **Restrict**: marginalize onto pair totals.
3. **Coarse solve**: advance the coarse RDME.
4. **Correct**: prolong the coarse correction $\delta^{2h} = \tilde{p}^{2h} - p^{2h}$ to the fine level.
5. **Post-smooth**: re-equilibrate within-pair distributions.

Only the correction δ^{2h}, not the full coarse distribution, is prolongated. This keeps the fine state space small: it grows only from the fraction of probability that moves during the coarse solve, not from the entire coarse support.

For linear birth–death–diffusion systems, the coarse operator closes exactly. For nonlinear systems such as the Schlögl model, Binomial within-pair equilibration gives explicit coarse propensities. The V-cycle is accurate in homogeneous bulk but fails at bistable interfaces, where the fine conditional distribution deviates sharply from Binomial. This failure mode leads naturally to an adaptive strategy: coarsen only where the local distribution is redundant, and retain fine resolution where the interface structure matters.

The adaptive extension is a persistent mixed-resolution FSP: fine and coarse pairs are co-evolved on a single mixed-resolution state space, the refinement region is updated adaptively, and the full fine grid is never materialized. Second-level quad-coarsening (merging two pair-coarse voxels) extends the hierarchy one level further, with a policy validation confirming that the adaptive criterion keeps this aggressive coarsening confined to spatially distant bulk pairs.

The language of restriction and prolongation is classical multigrid [1, 11], but the reduced operator is a probability generator rather than an elliptic discretization, and the prolongation is probabilistic rather than interpolatory. The present manuscript is focused on the one-dimensional single-species case to expose the main ideas clearly. Within that scope, the contributions are as follows.

1. We formulate pairwise spatial coarsening as a Kemeny–Snell aggregation of the RDME Markov chain [2, 7] and derive the exact Galerkin coarse generator.
2. We construct a two-level FAS-style V-cycle for the probability generator, prove exactness for linear networks, and characterize the Binomial closure used in nonlinear coarse propensities.
3. We identify the interface failure mode of uniform coarsening and introduce an adaptive mixed-resolution strategy that avoids coarsening such regions.
4. We validate second-level quad-coarsening on a twelve-voxel domain, including a policy audit showing that the adaptive criterion protects front-adjacent pairs from quad coarsening.

The intended regime is not large-scale bulk simulation, where trajectory-based methods remain more practical, but low-count spatial systems for which the distribution itself matters: interfaces, rare switching events, and localized metastable structures.

2 RDME and spatial aggregation

2.1 Fine-grid RDME

Let a one-dimensional domain be partitioned into K voxels of width h, and let

$$\boldsymbol{x} = (n_1, \ldots, n_K) \in \mathbb{Z}_{\geq 0}^K$$

denote the vector of molecule counts. The RDME defines a continuous-time Markov chain on $\mathbb{Z}_{\geq 0}^K$ with generator

$$A = A_{\mathrm{rxn}} + A_{\mathrm{diff}}.$$

The reaction part acts independently within each voxel, while the diffusion part transports molecules between neighboring voxels at rate $d = D/h^2$.

Given a truncated state space $\mathcal{S}_h \subset \mathbb{Z}_{\geq 0}^K$, FSP replaces the full generator by its restriction to $\mathcal{S}_h$ and evolves the projected distribution by matrix exponentiation [8]. In spatial problems, the size of $\mathcal{S}_h$ grows rapidly with K, and the fine-grid diffusion rate $d = D/h^2$ becomes large as $h \to 0$.

2.2 Pair aggregation as a Markov-chain lumping

Assume K is even and partition the fine grid into voxel pairs $\{1,2\}, \{3,4\}, \ldots, \{K-1, K\}$. Define the coarse variables

$$\bar{n}_j = n_{2j-1} + n_{2j}, \qquad j = 1, \ldots, K/2,$$

and let $\bar{\boldsymbol{x}} = (\bar{n}_1, \ldots, \bar{n}_{K/2})$. Restriction is exact marginalization over within-pair splits:

$$(\mathcal{R}p)(\bar{\boldsymbol{x}}) = \sum_{\boldsymbol{x}:\, n_{2j-1}+n_{2j}=\bar{n}_j \;\; \forall j} p(\boldsymbol{x}).$$

When intra-pair diffusion equilibrates rapidly relative to the remaining dynamics, the fine states compatible with the same $\bar{\boldsymbol{x}}$ mix toward a local stationary law. For identical neighboring subvoxels this law is Binomial:

$$\pi(m \mid \bar{n}_j) = \mathrm{Bin}(\bar{n}_j, \tfrac{1}{2})(m), \qquad m = 0, \ldots, \bar{n}_j.$$

The prolongation operator is the product of within-pair conditional laws,

$$(\mathcal{P}\bar{p})(\boldsymbol{x}) = \bar{p}(\bar{\boldsymbol{x}}) \prod_{j=1}^{K/2} \pi(n_{2j-1} \mid \bar{n}_j),$$

and the coarse generator is

$$\bar{A} = \mathcal{R}A\mathcal{P}.$$

This is the standard Kemeny–Snell form for aggregation of a Markov chain by fast local equilibration [2, 7, 10].

Remark 1. *The same operator template appears in quasi-steady-state reductions of well-mixed stochastic reaction networks, where fast reaction subnetworks are collapsed and the prolongation is defined by a local invariant distribution [3, 4, 9]. In the present setting, the fast process is spatial diffusion rather than a fast reaction subsystem.*

3 Two-level FAS-style V-cycle

3.1 Algorithm

Algorithm 1 states the two-level cycle precisely. The key design choices are:

- **Smoother**: intra-pair diffusion only. This drives within-pair distributions toward the Binomial equilibrium $\pi(\cdot \mid \bar{n}_j)$ without perturbing the pair totals $\bar{n}_j$. It is a genuine smoother for the fast spatial modes introduced by fine-grid diffusion.

Algorithm 1 Two-level FAS V-cycle for the RDME

Require: Fine distribution p^h, time step Δt, smoothing times $\tau_{\text{pre}}, \tau_{\text{post}}$

1: **Pre-smooth**: $\tilde{p}^h \leftarrow e^{A_{\text{intra}}\,\tau_{\text{pre}}} p^h$ {intra-pair diffusion only}

2: **Restrict**: $p^{2h} \leftarrow \mathcal{R}\,\tilde{p}^h$

3: **Coarse expand**: add FSP boundary shells to $\mathcal{S}_{2h}$

4: **Coarse solve**: $\tilde{p}^{2h} \leftarrow e^{\bar{A}\,\Delta t}\, p^{2h}$

5: **Correction**: $\delta^{2h} \leftarrow \tilde{p}^{2h} - p^{2h}$

6: **Prolong correction**: $p^h_{\text{new}} \leftarrow \tilde{p}^h + \mathcal{P}\,\delta^{2h}$

7: **Post-smooth**: $p^h_{\text{new}} \leftarrow e^{A_{\text{intra}}\,\tau_{\text{post}}} p^h_{\text{new}}$

Ensure: Updated fine distribution p^h_{new}

- **Correction, not substitution**: only the coarse correction $\delta^{2h} = \tilde{p}^{2h} - p^{2h}$ is prolongated and added to the pre-smoothed fine distribution. The fine state space grows only from the probability that moved during the coarse solve, not from the entire coarse support. This is the Full Approximation Scheme (FAS) applied to the probability generator.
- **Signed prolongation**: δ^{2h} is signed. To apply the (non-negative) prolongation operator, positive and negative parts are prolongated separately and then combined. Small numerically-negative probabilities arising from cancellation are clipped to zero.

3.2 Coarse operators and closure

3.2.1 Linear birth–death–diffusion systems

Consider a one-species birth–death–diffusion model with per-voxel reactions $\varnothing \xrightarrow{k_b} A$ and $A \xrightarrow{k_d} \varnothing$, and nearest-neighbor diffusion. Under pair aggregation:

- the coarse diffusion rate is $D/(2h)^2 = d/4$;
- the coarse birth rate doubles (one coarse voxel represents two fine voxels);
- the death rate remains first-order in the coarse count.

Since the generator is linear in the state, the aggregated operator closes exactly.

Proposition 1. *For the one-dimensional birth–death–diffusion RDME with pairwise spatial aggregation and Binomial prolongation, the Galerkin coarse generator $\bar{A} = \mathcal{R}A\mathcal{P}$ coincides with the exact RDME generator on the $K/2$-voxel coarse grid.*

Figure 1 confirms this numerically: the coarse and fine solutions are indistinguishable at the distribution level while the coarse state space is dramatically smaller.

3.2.2 Nonlinear Schlögl closure

We use the one-dimensional Schlögl model as the canonical nonlinear test problem. In each voxel,

$$\varnothing \xrightarrow{c_3} X, \qquad X \xrightarrow{c_4} \varnothing, \qquad 2X \xrightarrow{c_1} 3X, \qquad 3X \xrightarrow{c_2} 2X.$$

The coarse zeroth- and first-order propensities remain exact under pair aggregation. The autocatalytic and reverse reactions require conditional moments of the within-pair split; under the Binomial

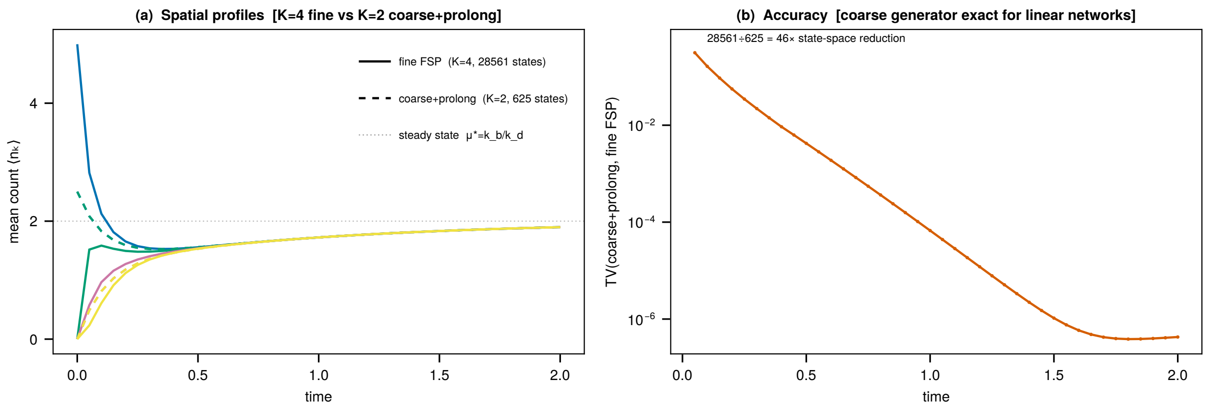

Figure 1: Linear birth–death–diffusion benchmark. The pair-aggregated coarse generator is exact for this linear system, and the coarse solution prolonged to the fine grid matches the fine FSP solution while using far fewer states.

model $n_{2j-1} \mid \bar{n}_j \sim \mathrm{Bin}(\bar{n}_j, \frac{1}{2})$:

$$a^{\text{coarse}}_{\text{auto}}(\bar{n}_j) = c_1 \, \frac{\bar{n}_j(\bar{n}_j - 1)}{4},$$
$$a^{\text{coarse}}_{\text{rev}}(\bar{n}_j) = c_2 \, \frac{\bar{n}_j(\bar{n}_j - 1)(\bar{n}_j - 2)}{24}.$$

The closure is accurate when the within-pair conditional is close to the fast-diffusion equilibrium. Figure 2 compares exact conditional factorial moments from a fine FSP solution against the Binomial predictions: a bulk pair is near symmetric equilibrium with small closure error, while an interface pair is highly asymmetric and the closure error is order-one.

4 V-cycle results: nonlinear Schlögl front

Figure 3 shows the V-cycle applied to the Schlögl model on an eight-voxel grid with a bistable front initial condition. Voxel means from the V-cycle solution are compared against a direct fine-grid FSP reference. The V-cycle tracks the reference accurately: the front position and amplitude are captured without materializing the full fine-grid state space at each step.

Figure 4 quantifies accuracy more precisely. Total variation distance between the V-cycle prolongated distribution and the fine reference is small and stable, confirming that the coarse correction mechanism is propagating the relevant fine-level information faithfully. Figure 5 shows the stiffness reduction: coarsening replaces the fine diffusion rate D/h^2 by $D/(2h)^2$, doubling the largest accepted time step at fixed distribution error.

5 Failure of uniform coarsening at fronts

5.1 The interface failure mode

If an interface lies inside an aggregated pair, then the coarse variable $\bar{n}_j$ loses the orientation of the front. In the Schlögl front regime, the two fine states (n_ℓ, n_h) and (n_h, n_ℓ) have the same coarse

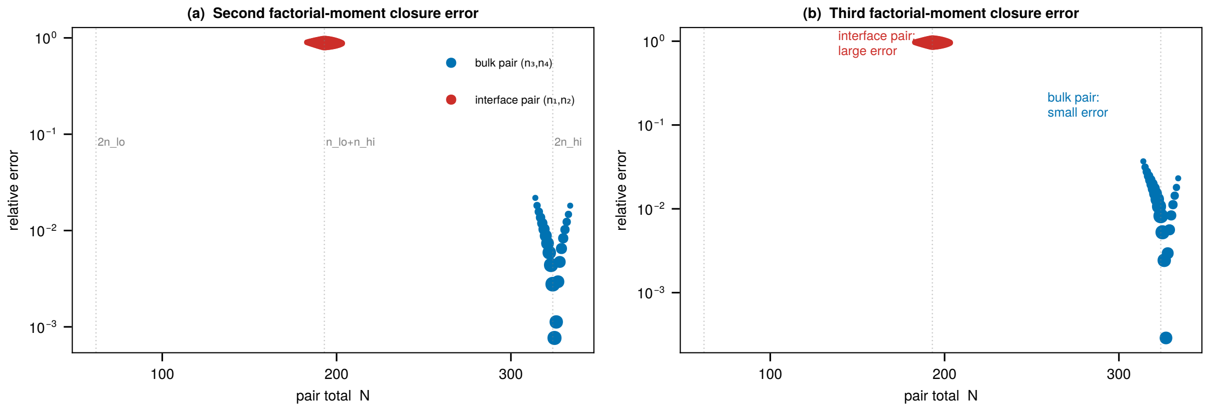

Figure 2: Conditional-moment closure for the Schlögl model. Binomial closure is accurate for a homogeneous bulk pair but fails sharply at a bistable interface. This is the structural reason the V-cycle and the adaptive method retain interface pairs at fine resolution.

sum $n_\ell + n_h$ but represent distinct spatial configurations. Uniform coarsening therefore destroys exactly the information needed to resolve the interface distribution.

5.2 Refinement diagnostics

The adaptive criterion combines three diagnostics defined from the current distribution.

1. **Within-pair asymmetry**

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}, \qquad \mu_k = \mathbb{E}[n_k].$$

2. **Inter-pair gradient**

$$\beta_j = \frac{|\mu_{2j} - \mu_{2j+1}|}{\mu_{2j} + \mu_{2j+1}}.$$

3. **Per-molecule flux**

$$\varphi_j = \frac{d\,\mu_{2j-1}/\max(1, \mu_{2j-1})}{\bar{\varphi}_{\text{domain}}},$$

guarding low-count transport bottlenecks.

The effective asymmetry for front detection is

$$\alpha_j^{\text{eff}} = \max\{\alpha_j, \beta_{j-1}, \beta_j\}.$$

Hysteresis thresholds $(\alpha_{\text{lo}}, \alpha_{\text{hi}})$ prevent rapid mask oscillation. Figure 6 illustrates three representative scenarios: a front inside a pair, a front on a pair boundary, and a homogeneous region.

5.3 Localized refinement region

Rather than allowing arbitrary disconnected fine pairs, we define a contiguous refinement region. Pairs flagged by the criterion are collected, the smallest interval containing them is formed, and an optional halo of neighboring pairs is added. This localized refinement region is aligned with the spatial structure of fronts and simplifies repartitioning.

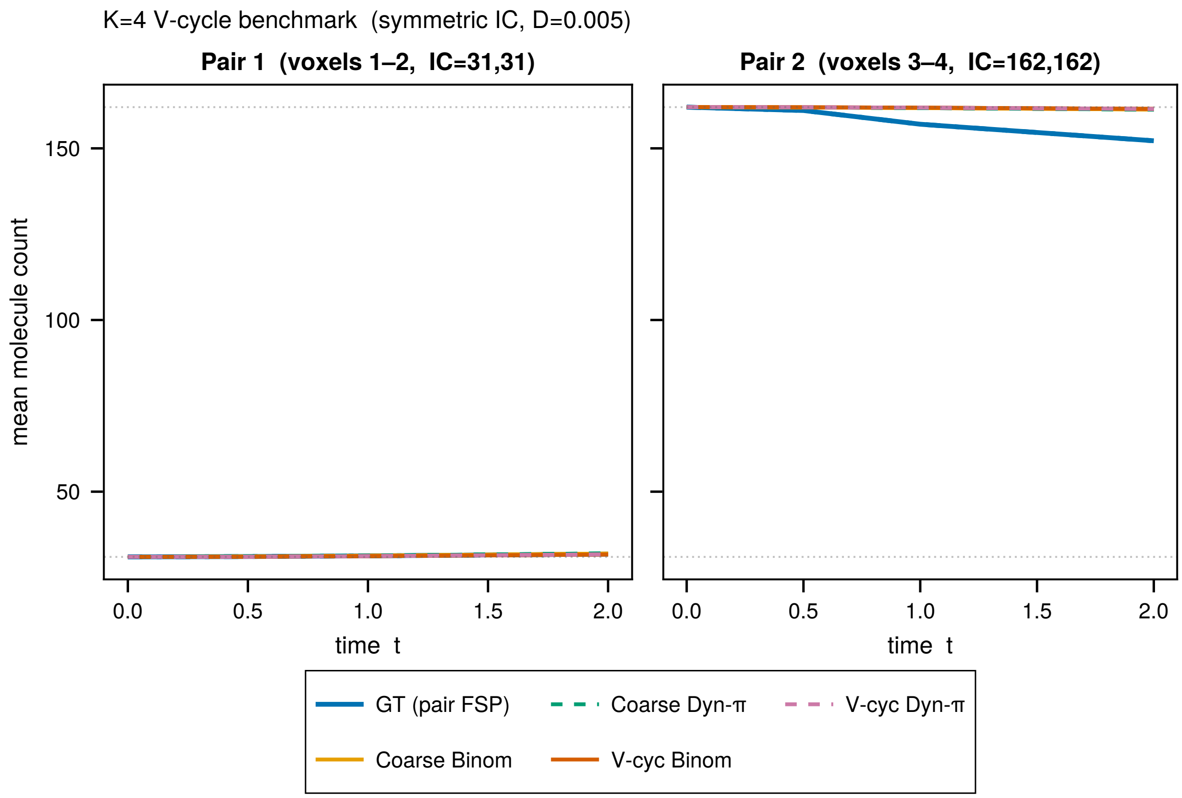

Figure 3: V-cycle voxel means versus fine FSP reference on the $K = 8$ Schlögl bistable front. The two-level FAS cycle tracks the reference spatial profile accurately throughout the evolution.

6 Adaptive mixed-resolution extension

6.1 Mixed-resolution RDME

Once the refinement region is selected, the state lives on a mixed-resolution space $\mathcal{S}^M \subset \mathbb{Z}_{\geq 0}^{K_M}$, where each fine pair contributes two coordinates and each coarse pair contributes one. The mixed generator is assembled directly:

- fine pairs use the original voxel-level reactions and intra-pair diffusion,
- coarse pairs use the Binomial-closed coarse propensities,
- cross-pair diffusion uses first-moment boundary closure when one side is coarse.

Interface pairs are never coarsened, so their within-pair distribution is never approximated by a Binomial law.

6.2 Persistent mixed-state evolution

Rather than taking one mixed step and prolongating back to the full fine grid, we evolve directly on the mixed-resolution state space:

1. evaluate the refinement criterion when required,

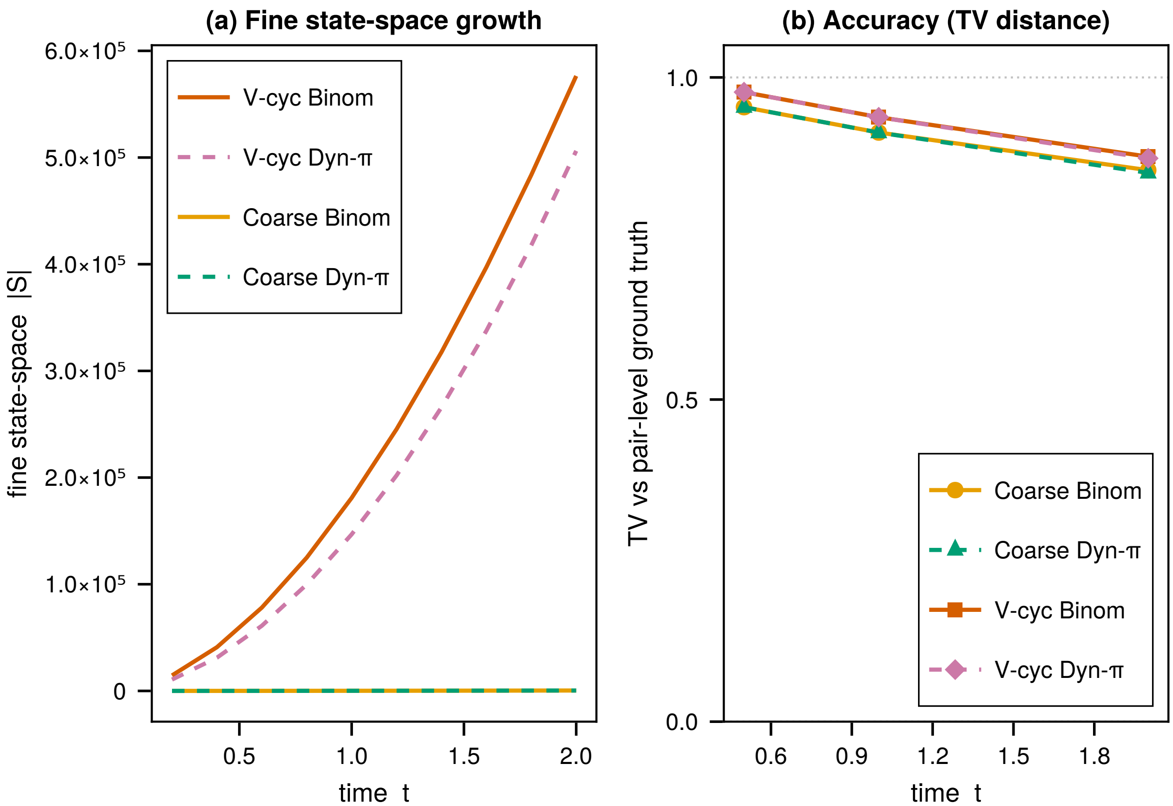

Figure 4: V-cycle accuracy on the $K = 8$ Schlögl front. Total variation distance between the V-cycle prolongated distribution and the direct fine FSP reference remains small throughout the run, confirming coarse correction quality.

2. update the partition only if the localized refinement region changes,
3. adapt the current mixed state by pairwise marginalization or Binomial splitting,
4. expand the mixed FSP state space,
5. solve the mixed RDME over one time step,
6. prune low-probability states.

Fine-grid reconstruction is performed only when a fine observable is requested. A pair-space center-of-mass triggers a full repartition only when the front drifts past a threshold.

6.3 Adaptive front resolution

Figure 7 compares direct FSP, uniform coarsening, and adaptive coarsening on a $K = 4$ Schlögl front problem. The adaptive method retains the front pair at fine resolution and coarsens the homogeneous pair, reducing the approximation error substantially relative to uniform coarsening. Figure 8 shows the corresponding time-dependent mask.

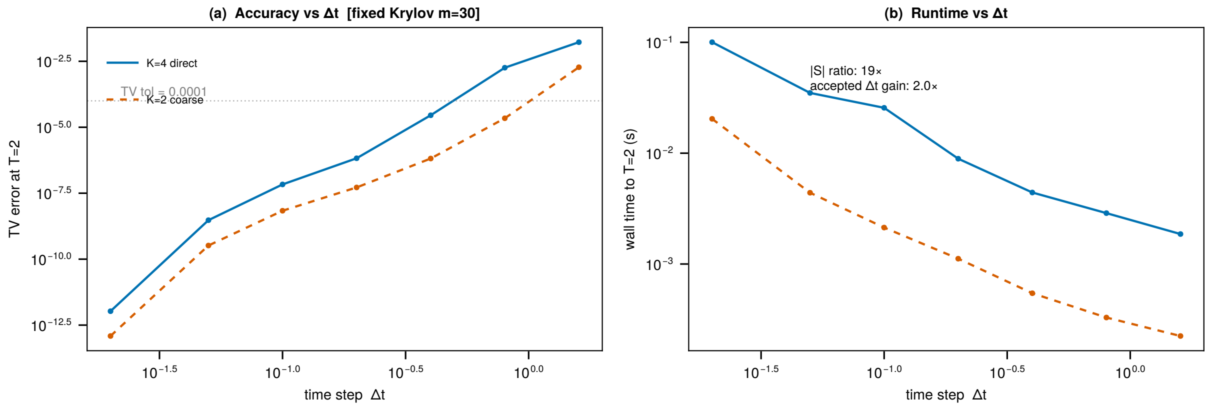

Figure 5: Empirical stiffness benchmark. Coarsening reduces the diffusion stiffness and permits larger accepted time steps at fixed distributional error.

6.4 Persistent mixed-resolution evolution on an eight-voxel front

Figure 9 shows the main large-grid result. A Schlögl front on a $K = 8$ grid is evolved directly on the mixed-resolution state space. The coarse marginal $\bar{n}_2 = n_3 + n_4$ is insufficient to distinguish the two fine interface configurations, while fine resolution inside the active pair preserves the interface distribution.

Figure 10 compares state-space growth in direct and adaptive coarsened FSP solves, confirming dramatic reduction under coarsening.

6.5 FSP versus trajectory sampling on the same mixed model

Figure 11 compares direct FSP on a three-dimensional mixed CME against stochastic simulation of the same mixed generator. The TV distance between SSA and FSP decays at the expected $N^{-1/2}$ Monte Carlo rate. FSP returns the full interface distribution directly; trajectory sampling resolves it only gradually.

7 Second-level coarsening and policy validation

7.1 Second-level (quad) coarsening

The pair-aggregation scheme extends to a second coarsening level by merging two adjacent level-1 pairs into a single *quad* variable $\tilde{n} = \bar{n}_j + \bar{n}_{j+1}$, representing four fine voxels with $r = 4$ sub-voxels. The Binomial closure at this level gives

$$a^{(2)}_{\text{auto}}(\tilde{n}) = c_1 \frac{\tilde{n}(\tilde{n}-1)}{8},$$
$$a^{(2)}_{\text{rev}}(\tilde{n}) = c_2 \frac{\tilde{n}(\tilde{n}-1)(\tilde{n}-2)}{96}.$$

Two adjacent level-1 pairs are promoted to a quad only when both satisfy $\alpha_j^{\text{eff}} < \alpha_{\text{lo2}}$ with $\alpha_{\text{lo2}} \ll \alpha_{\text{lo}}$. The set of level-2 pairs is therefore always a subset of the level-1 bulk region, well separated from any front.

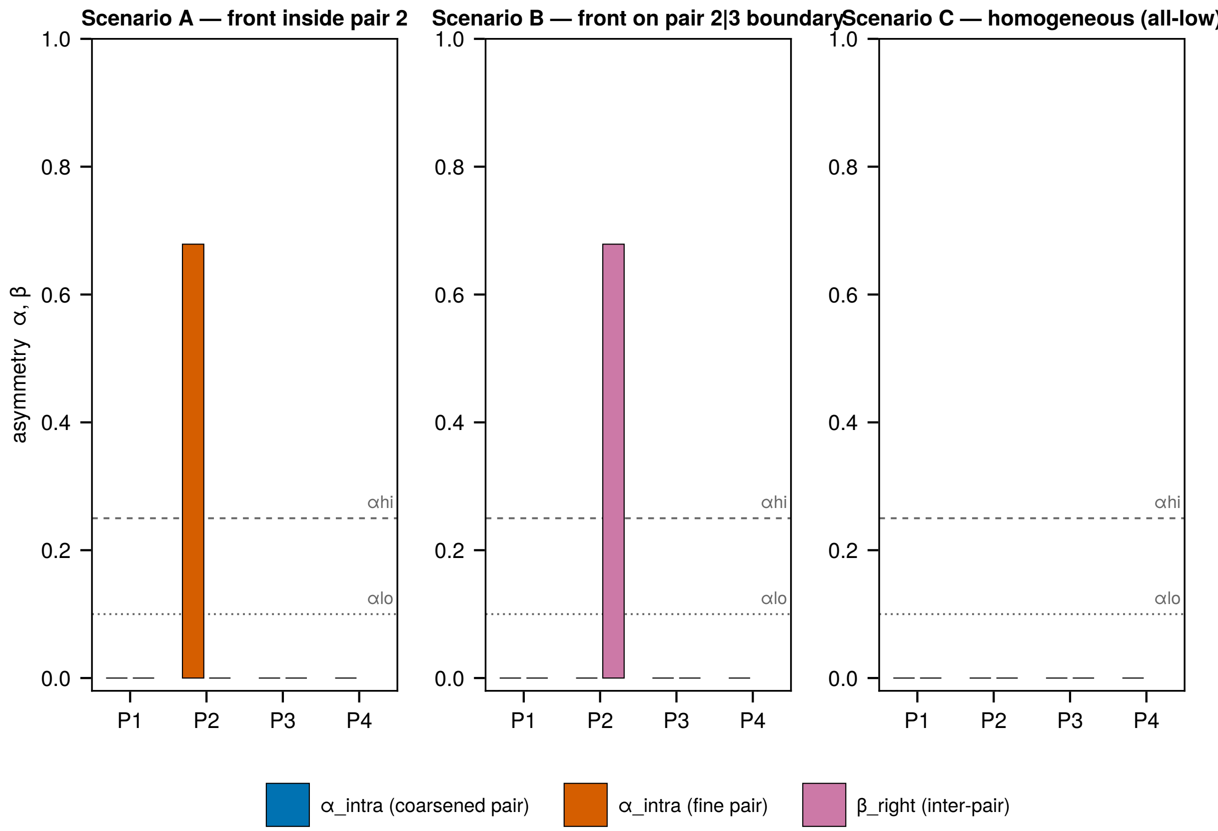

Figure 6: Adaptive refinement diagnostics on representative $K = 8$ Schlögl states. The within-pair asymmetry α detects fronts inside a pair, while the inter-pair gradient β detects fronts aligned with a pair boundary.

7.2 Isolation test

With $K = 4$ voxels (two pairs), a direct fine FSP is tractable and provides a reference. Table 1 shows the TV between the reference projected onto pair-sum variables and the corresponding coarse distribution, at $t = 0.5$ with voxel spacing $h = 1/12$ (matching $K = 12$).

The level-1-vs-level-2 TV (rightmost column) is the intrinsic cost of the second-level Binomial closure: 0.036–0.079 in bulk, 0.266 at the front — seven times larger.

7.3 Policy validation

Table 2 reports per-pair diagnostics at $t = 0.5$ in the $K = 12$ adaptive run. The effective asymmetry α_j^{eff} governs level assignment.

The inter-pair gradient β is the critical protection mechanism. Pairs 2 and 4 have near-zero internal asymmetry but inherit large α^{eff} from the front at pair 3 through β, blocking level-2 promotion. Only pairs 5 and 6, separated from the front by two L1 guards, satisfy $\alpha^{\mathrm{eff}} < \alpha_{\mathrm{lo2}}$ and are promoted.

The coarse-marginal TV between the level-1 and level-2 runs at $t = 0.5$ is 0.034, consistent with the bulk regime (0.036) in the isolation test. This confirms the observed approximation cost is the

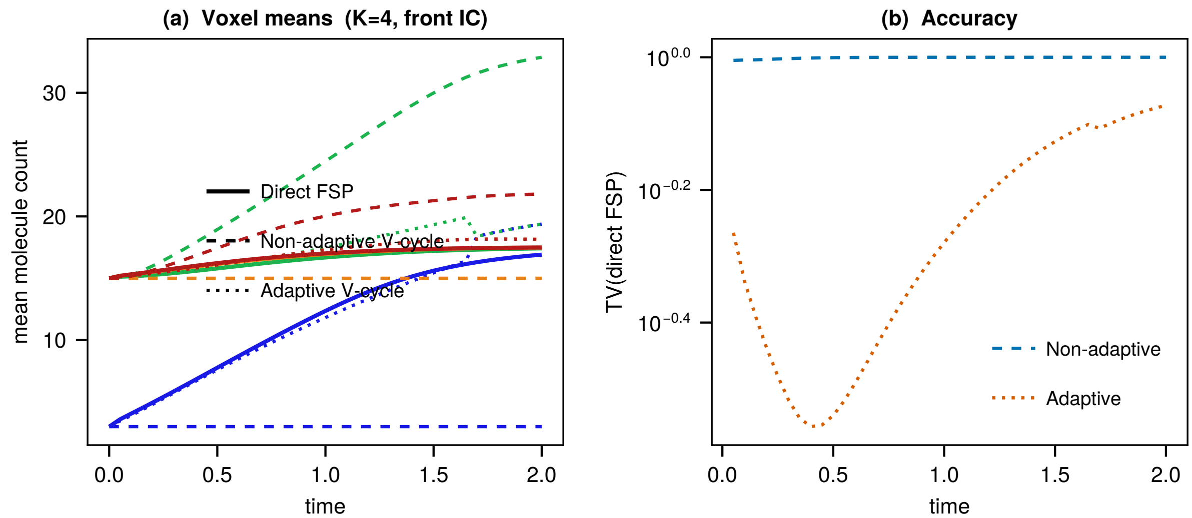

Figure 7: Accuracy of the adaptive two-level method on a $K = 4$ Schlögl front problem. The adaptive method tracks the direct FSP reference substantially better than uniform coarsening.

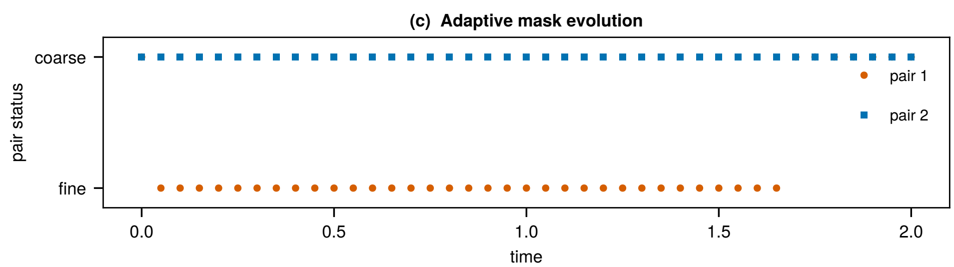

Figure 8: Time-resolved adaptive mask on the $K = 4$ Schlögl front. The front pair remains at fine resolution while the homogeneous pair is coarsened.

intrinsic cost of the second-level closure in a homogeneous bulk region, not a front-contamination artifact.

8 Discussion

The present results support four main conclusions.

First, spatial multigrid for the RDME can be formulated as a genuine probability-generator reduction rather than a heuristic coarsening. Restriction and prolongation arise naturally from local diffusion equilibration, the coarse generator is exact in the linear setting, and the two-level cycle has the complete FAS structure: pre-smooth, restrict, coarse solve, correct, post-smooth.

Second, the smoother is not an arbitrary component. Intra-pair diffusion acts precisely on the fast modes introduced by fine-grid spatial discretization. After pre-smoothing, within-pair distributions are near Binomial and the restriction operator $\mathcal{R}$ is near-exact. This is what makes the correction $\delta^{2h} = \tilde{p}^{2h} - p^{2h}$ small and the V-cycle effective.

Third, the main obstacle in nonlinear bistable systems is not coarsening itself but coarsening

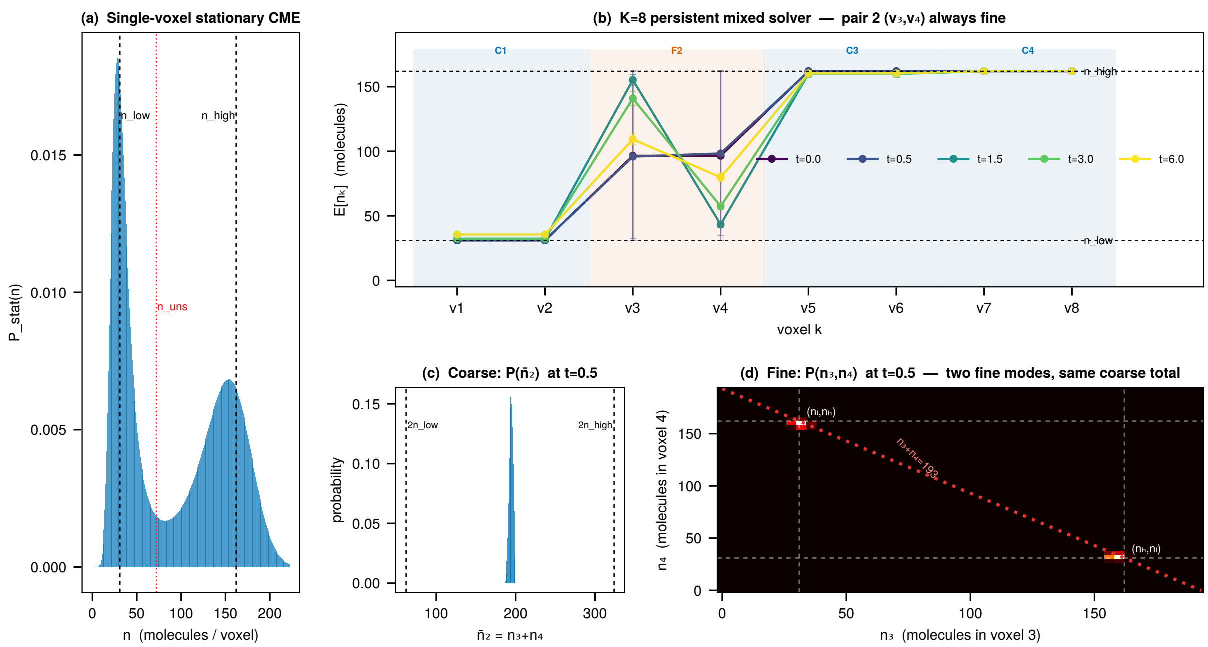

Figure 9: $K = 8$ mixed-resolution Schlögl front. Pair 2 is retained at fine resolution while the surrounding bulk is coarsened. The coarse variable $\bar{n}_2$ cannot distinguish the two fine interface orientations, whereas the fine pair distribution resolves them explicitly.

the wrong region. Uniform Binomial prolongation fails at interfaces because the fine conditional is not near the local equilibrium. The adaptive strategy resolves this by keeping the active front at fine resolution and applying coarse closure only in bulk regions where it is appropriate.

Fourth, the second-level coarsening experiment provides the first evidence that the policy extends beyond the first coarsening level. The effective asymmetry α_j^{eff}—including the inter-pair gradient β rather than only the internal α_j—produces a protective halo around the front. Pairs immediately adjacent to the interface have large α_j^{eff} even when their internal asymmetry is negligible, because the front gradient leaks into them through β. In the $K = 12$ test, level-2 coarsening is confined to regions spatially distant from the active front, where the bulk Binomial closure at $r = 4$ introduces 3–8% TV error. Whether the same gradient criterion remains sufficient beyond level 2, or for more complex front geometries, is an open question.

The present work has limitations. The most important unresolved issue is state growth in the mixed FSP when several coarse directions become dynamically broad; systematic per-coordinate truncation control is a natural next step. Richer closures for coarse reaction terms at higher coarsening levels, and extension to multi-species and two-dimensional domains, remain open.

9 Conclusion

We have presented a two-level multigrid FSP method for the RDME based on spatial aggregation of adjacent voxels. The aggregation is a Kemeny–Snell lumping driven by fast intra-pair diffusion, yielding a probabilistic restriction–prolongation pair and a Galerkin coarse generator. The two-level cycle uses intra-pair diffusion as the smoother, exact marginalization as restriction, a coarse

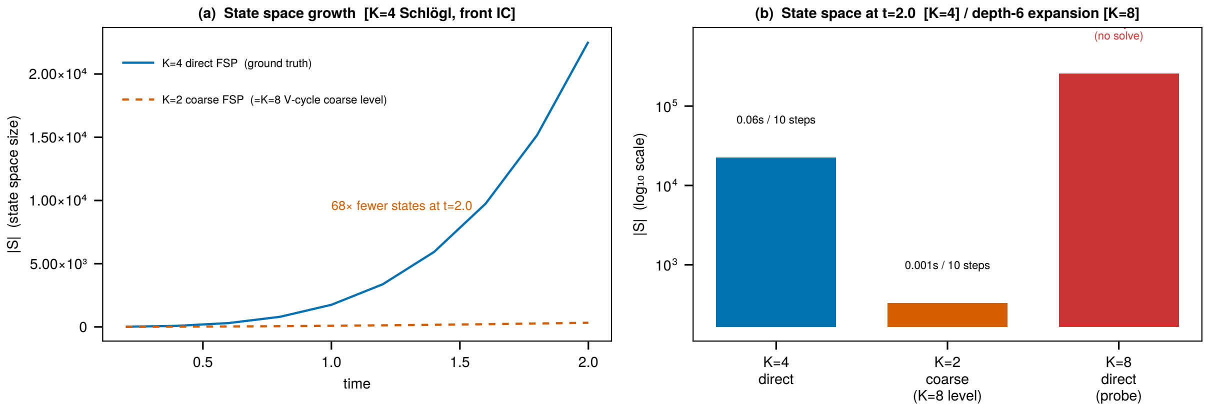

Figure 10: State-space reduction under adaptive coarsening on a Schlögl front problem.

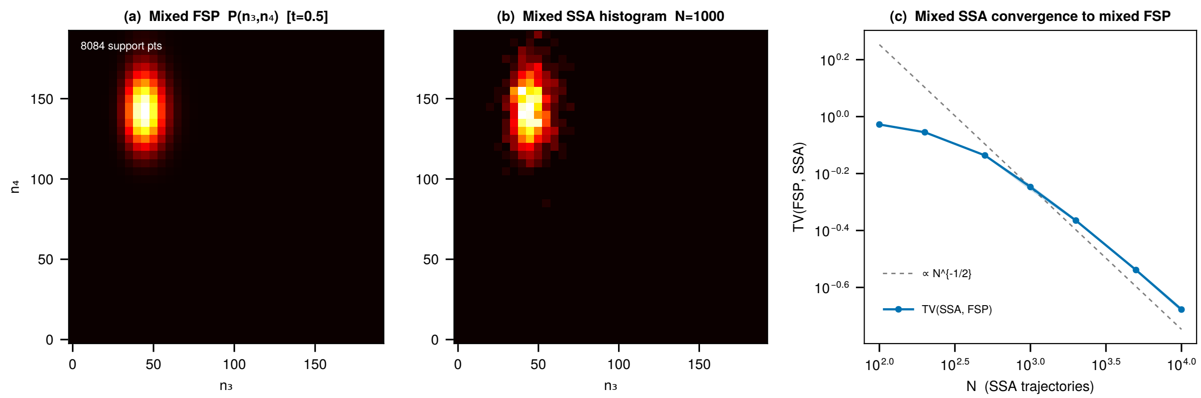

Figure 11: Mixed FSP versus mixed SSA on the same reduced model. FSP directly resolves the interface distribution; SSA converges at the expected Monte Carlo rate.

RDME solve, and prolongation of the signed coarse correction. This construction is exact for linear birth–death–diffusion systems and provides a tractable FAS-style cycle for nonlinear systems.

In bistable Schlögl front problems, the cycle is accurate in homogeneous bulk and fails at interfaces; this motivates adaptive mixed-resolution evolution, which retains fine resolution only where the distribution contains nonredundant spatial structure. Second-level quad-coarsening extends the hierarchy one level, and the $K = 12$ policy audit shows empirically that the adaptive α^{eff} criterion confines it to bulk pairs in this setting, so the observed coarse-marginal TV is the intrinsic cost of the bulk closure rather than a front artifact. The resulting method resolves interface distributions, reduces state-space dimension, and provides stiffness relief, all within the FSP framework.

References

[1] William L Briggs, Van Emden Henson, and Steve F McCormick. *A Multigrid Tutorial*. SIAM, Philadelphia, 2nd edition, 2000.

Table 1: $K = 4$ level-2 isolation test at $t = 0.5$. All TV values compare distributions on the same coarsened variable. The front scenario measures the catastrophic case excluded by the adaptive policy.

Scenario	$\mathrm{TV}(p^{\mathrm{fine}}\!\downarrow, p^{L1})$	$\mathrm{TV}(p^{\mathrm{fine}}\!\downarrow, p^{L2})$	$\mathrm{TV}(p^{L1}\!\downarrow, p^{L2})$
Both pairs HIGH (bulk)	0.303	0.184	0.079
Both pairs LOW (bulk)	0.204	0.117	0.036
Pair 1 LOW, pair 2 HIGH (front)	0.451	0.293	0.266

Table 2: Per-pair diagnostics at $t = 0.5$ in the $K = 12$ adaptive run. Thresholds: $\alpha_{\mathrm{lo}} = 0.10$, $\alpha_{\mathrm{hi}} = 0.25$, $\alpha_{\mathrm{lo2}} = 0.03$.

Pair	Level	μ_L	μ_R	α_j	α_j^{eff}	Status
1	L1	31.0	31.0	0.000	0.003	bulk
2	L1	31.1	31.1	0.000	0.513	**front halo** ($\beta_{2\to3} = 0.51 \gg \alpha_{\mathrm{lo}}$)
3	L0	96.6	97.3	0.003	0.513	**front** ($> \alpha_{\mathrm{hi}}$, forced fine)
4	L1	161.8	161.8	0.000	0.249	**front halo** ($\beta_{3\to4} = 0.25 > \alpha_{\mathrm{lo}}$)
5	L2	162.0	162.0	0.000	0.001	bulk ($< \alpha_{\mathrm{lo2}}$, promoted)
6	L2	162.0	162.0	0.000	0.000	bulk ($< \alpha_{\mathrm{lo2}}$, promoted)

[2] Peter Buchholz. Exact and ordinary lumpability in finite markov chains. *Journal of Applied Probability*, 31(1):59–75, 1994.

[3] Yang Cao and Linda Petzold. Slow-scale tau-leaping method. *Computer Methods in Applied Mechanics and Engineering*, 197(43):3472–3479, 2008. ISSN 0045-7825. doi: https://doi.org/10.1016/j.cma.2008.02.024. URL `https://www.sciencedirect.com/science/article/pii/S0045782508000959`. Stochastic Modeling of Multiscale and Multiphysics Problems.

[4] Youfang Cao, Jie Liang, Derrick Pettigrew, and S Kou. Adaptive aggregation of state-space models for the chemical master equation. *Multiscale Modeling & Simulation*, 14(1):28–48, 2016.

[5] Radek Erban, Jonathan Chapman, and Philip Maini. A practical guide to stochastic simulations of reaction-diffusion processes. *arXiv preprint arXiv:0704.1908*, 2007.

[6] Daniel T Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977.

[7] John G Kemeny and J Laurie Snell. *Finite Markov Chains*. Van Nostrand, Princeton, NJ, 1960.

[8] Brian Munsky and Mustafa Khammash. Finite state projection for the solution of the chemical master equation. *The Journal of Chemical Physics*, 124(4):044104, 2006.

[9] Slaven Peleš, Brian Munsky, and Mustafa Khammash. Reduction and solution of the chemical master equation using time scale separation and finite state projection. *The Journal of Chemical Physics*, 125(20):204104, 11 2006. ISSN 0021-9606. doi: 10.1063/1.2397685. URL `https://doi.org/10.1063/1.2397685`.

[10] Herbert A Simon and Albert Ando. Aggregation of variables in dynamic systems. *Econometrica*, 29(2):111–138, 1961.

[11] Ulrich Trottenberg, Cornelis W Oosterlee, and Anton Schüller. *Multigrid*. Academic Press, London, 2001.